# 在不使用 Upgrade Advisor 进行升级之前，应验证哪些内容？

ONTAP 9

NetApp
February 24, 2022

This PDF was generated from https://docs.netapp.com/zh-cn/ontap/upgrade/task_what_to_verify_before_upgrade.html on February 24, 2022. Always check docs.netapp.com for the latest.

# 目录

在不使用 Upgrade Advisor 进行升级之前，应验证哪些内容？ 1
升级前要验证的内容 1
验证集群升级限制 1
验证当前集群活动 1
如何在 ONTAP 升级期间更新固件 4

# 在不使用 Upgrade Advisor 进行升级之前，应验证哪些内容？

## 升级前要验证的内容

如果不使用 "Active IQ" Upgrade Advisor 要规划升级，您应在升级之前验证集群升级限制和集群活动。

## 验证集群升级限制

如果不使用 "Active IQ" Upgrade Advisor ，则需要验证集群是否未超过平台系统限制。除了平台系统限制之外， SAN 还具有一些限制，您还应进行验证。

1. 验证集群是否未超过您平台的系统限制。

   "NetApp Hardware Universe"

2. 如果您的集群配置了 SAN ，请验证它是否未超过 FC ， FCoE 和 iSCSI 的配置限制。

   "NetApp Hardware Universe"

3. 确定 CPU 和磁盘利用率： `node run -node node_name -command sysstat -c 10 -x 3`

   您应监控 CPU 和磁盘利用率 30 秒。对于报告的所有 10 个测量值， "CPU* " 和 "* 磁盘利用率 * " 列中的值不应超过 50% 。在升级完成之前，不应向集群添加额外负载。注意： CPU 和磁盘利用率可能会在您的环境中的不同时间有所不同。因此，最好在预期升级窗口的时间段内检查 CPU 和磁盘利用率。

## 验证当前集群活动

如果不使用 "Active IQ" Upgrade Advisor ，在升级之前，您应手动验证是否未运行任何作业，以及是否终止任何不持续可用的 CIFS 会话。

### 确认没有正在运行的作业

在升级 ONTAP 软件之前，您必须验证集群作业的状态。如果存在任何聚合，卷， NDMP （转储或还原）或 Snapshot 作业（例如创建，删除，移动，修改，复制， 和挂载作业）正在运行或排队，您必须允许作业成功完成或停止已排队的条目。

1. 查看任何正在运行或已排队的聚合，卷或 Snapshot 作业的列表： `job show`

```
cluster1::> job show
                            Owning
Job ID Name                 Vserver    Node           State
------ -------------------- ---------- -------------- ----------
8629   Vol Reaper           cluster1   -              Queued
       Description: Vol Reaper Job
8630   Certificate Expiry Check
                            cluster1   -              Queued
       Description: Certificate Expiry Check
.
.
.
```

2. 如果有任何正在运行的作业，请允许它们成功完成。
3. 删除任何已排队的聚合，卷或 Snapshot 副本作业： `job delete -id job_id`

```
cluster1::> job delete -id 8629
```

4. 确认没有聚合，卷或 Snapshot 作业正在运行或已排队： `job show`

   在此示例中，已删除所有正在运行和已排队的作业：

```
cluster1::> job show
                            Owning
Job ID Name                 Vserver    Node           State
------ -------------------- ---------- -------------- ----------
9944   SnapMirrorDaemon_7_2147484678
                            cluster1   node1          Dormant
       Description: Snapmirror Daemon for 7_2147484678
18377  SnapMirror Service Job
                            cluster1   node0          Dormant
       Description: SnapMirror Service Job
2 entries were displayed
```

## 确定应终止的活动 CIFS 会话

在升级 ONTAP 软件之前，您应确定并正常终止所有不持续可用的 CIFS 会话。

持续可用的 CIFS 共享由 Hyper-V 或 Microsoft SQL Server 客户端使用 SMB 3.0 协议访问，无需在升级之前终止。

1. 确定任何已建立但不持续可用的 CIFS 会话： `vserver cifs session show -continuously -available Yes -instance`

此命令可显示有关任何不具有持续可用性的 CIFS 会话的详细信息。在继续 ONTAP 升级之前，您应先终止它们。

```
cluster1::> vserver cifs session show -continuously-available Yes
-instance

                        Node: node1
                     Vserver: vs1
                  Session ID: 1
                Connection ID: 4160072788
Incoming Data LIF IP Address: 198.51.100.5
      Workstation IP address: 203.0.113.20
    Authentication Mechanism: NTLMv2
                Windows User: CIFSLAB\user1
                   UNIX User: nobody
                 Open Shares: 1
                  Open Files: 2
                  Open Other: 0
              Connected Time: 8m 39s
                   Idle Time: 7m 45s
            Protocol Version: SMB2_1
      Continuously Available: No
1 entry was displayed.
```

2. 如有必要，请确定为您标识的每个 CIFS 会话打开的文件： `vserver cifs session file show -session-id session_ID`

```
cluster1::> vserver cifs session file show -session-id 1

Node:       node1
Vserver:    vs1
Connection: 4160072788
Session:    1
File    File      Open Hosting
Continuously
ID      Type      Mode Volume          Share                 Available
------- --------- ---- --------------- ---------------------
------------
1       Regular   rw   vol10           homedirshare          No
Path: \TestDocument.docx
2       Regular   rw   vol10           homedirshare          No
Path: \file1.txt
2 entries were displayed.
```

# 如何在 ONTAP 升级期间更新固件

由于升级 ONTAP 包括升级固件，因此您无需手动更新固件。执行 ONTAP 升级时，ONTAP 升级软件包附带的集群固件将复制到每个节点的启动设备，并自动安装新固件。

如果集群中的固件版本早于 ONTAP 升级软件包捆绑的固件，则会自动更新以下组件的固件：

- 系统和诊断：
  - BIOS
  - Flash Cache
  - 服务处理器（ SP ）
- Disk
- 磁盘架

如果需要，您还可以在 ONTAP 升级之间手动更新固件。

**Copyright Information**

Copyright © 2022 NetApp, Inc. All rights reserved. Printed in the U.S. No part of this document covered by copyright may be reproduced in any form or by any means-graphic, electronic, or mechanical, including photocopying, recording, taping, or storage in an electronic retrieval system- without prior written permission of the copyright owner.

Software derived from copyrighted NetApp material is subject to the following license and disclaimer:

THIS SOFTWARE IS PROVIDED BY NETAPP “AS IS” AND WITHOUT ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE, WHICH ARE HEREBY DISCLAIMED. IN NO EVENT SHALL NETAPP BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.

NetApp reserves the right to change any products described herein at any time, and without notice. NetApp assumes no responsibility or liability arising from the use of products described herein, except as expressly agreed to in writing by NetApp. The use or purchase of this product does not convey a license under any patent rights, trademark rights, or any other intellectual property rights of NetApp.

The product described in this manual may be protected by one or more U.S. patents, foreign patents, or pending applications.

RESTRICTED RIGHTS LEGEND: Use, duplication, or disclosure by the government is subject to restrictions as set forth in subparagraph (c)(1)(ii) of the Rights in Technical Data and Computer Software clause at DFARS 252.277-7103 (October 1988) and FAR 52-227-19 (June 1987).

**Trademark Information**

NETAPP, the NETAPP logo, and the marks listed at http://www.netapp.com/TM are trademarks of NetApp, Inc. Other company and product names may be trademarks of their respective owners.